Making our world more productive

Briefing

Industrial metaverse: Concept and use cases

Long version

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2030 vision – industrial metaverse

An ordinary day at a Linde site in 2030

An air separation unit production engineer and her colleague, an onsite technician, talk as a truck-filling robot is directed to fill liquid nitrogen into a bulk truck. Suddenly, her artificial intelligence (AI) assistant appears as a hologram on the edge of the site, waving her over to confirm the AI-generated production plan of the day and to remind her that it's time to meet with the SHEQ-team for another project. She walks over to the plant's office, puts on her virtual reality (VR) headset, and finds herself in the lobby of the virtual Linde meeting room.

Her AI assistant reappears to direct her towards the virtual conference room where her meeting will be held. Upon her entry, the room transforms to look like a site (another air separation unit) using a real-time feed from a drone onsite. Piping isometric drawings from Linde Engineering are cloud-based laid over the live footage, and the production engineer and other meeting participants begin to assess the project work. Upon approval by a SHEQ expert, she sends her assistant to submit the applications for the next round of work permits. Then she pops off her headset, and heads back out to the truck filling[1].

Even if at an early stage today, the metaverse and underlying technologies will transform entire enterprise landscapes over the next decade.

Executive Summary

What is the metaverse? The metaverse is a spectrum of different digitally enhanced worlds converging physical and digital environments and therefore involving and connecting a wide range of technologies. 3D immersive environments will be one substantial part of the metaverse, but digital environments will also be represented by non-3D digital twins, i. e. by dynamic digital representations of real-world objects in real time. The metaverse is regarded as next evolution of the internet.



[1] Accenture (2022): Technology Vision 2022, Meet Me in the Metaverse, adapted to the Linde value chain

Definition

Historical background

Influenced by pioneering advancements in virtual reality and interface work by Autodesk's John Walker and VPL Research's Jaron Lanier, the American science-fiction author Neal Stephenson published the novel Snow Crash in 1992. In his book he described the online environment as a real place to its users in which they socialized, conducted business, and were entertained. The neologism "Metaverse" he had forged as a combination of the words "meta" and "universe" and used to describe this environment was well received by tech-enthusiasts of that time.[2]

> "Hiro is approaching the Street. It is the Broadway, the Champs Èlysées of the Metaverse. [...] It does not really exist. But right now, millions of people are walking up and down it."
>
> *Quote from Snow Crash*

Led by Ivan Sutherland's ideas, who in 1965 had already stated to: "make that virtual world in the window look so real, sound real, feel real, and respond realistically to the viewer's actions", a whole scientific community had formed around the vision of crafting fully immersive systems that place humans in computer generated models of the real world.[3] Publications of that time utilized keywords such as *Virtual Reality*, *Synthetic Experience*, *Virtual Worlds*, *Artificial Worlds*, *Artificial Reality*, *Virtual Environments* and finally from 1992 on: *Metaverse*.

At the same time, a paradigm shift initiated by Mark Weiser, the CTO of Xerox PARC lead to the birth of a new research discipline. Instead of virtual reality, he viewed Ubiquitous Computing (Ubicomp) as the future influence of technology on daily life: "Next comes ubiquitous computing, or the age of calm technology, when technology recedes into the background of our lives." In opposition to the virtual reality approach, Ubicomp started from social science insights, avoided personal computers yet envisioned multiple displays, MEMS sensors and small, networked computers throughout the real world. At the scale of doorhandles and light switches, a person would not own one computer, but rather use multiple ones, without spending cognitive energy throughout their day. Ubicomp envisioned not the computer, but the human in the centre. Subsequently, Cyber Physical Systems and the Industry 4.0 vision were built based on Ubicomp's infrastructure.

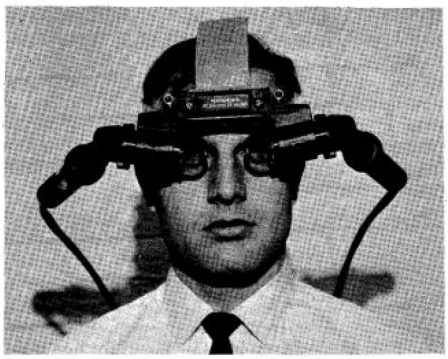


Figure 1: Sutherland's VR wearables from 1968

[2] Ondrejka (2005): Escaping the Gilded Cage: User Created Content and Building the Metaverse
[3] Mazuryk & Gervautz (1996): Virtual Reality – History, Applications, Technology and Future

Current state of the metaverse

Even 30 years later, the development of today's metaverse is still in its infancy and consequently constantly evolving.[4] Therefore, there is no singular all-encompassing definition people can turn to. In this paper it is regarded as blending of the aspirations of the above-mentioned disciplines Virtual Reality and Ubiquitous Computing. It will, at its core carry digital transformation into every aspect of our physical lives by creating a gigantic, immersive, persistent, and shared realm.[5]

A widely spread view is: the metaverse is where the internet becomes something consumers are immersed in, rather than something they just view

For this, the real and analogue world will be continually measured through various sensors to be interpreted by computers. Prevailing devices will know the boundaries of the room you are in, where you put your coffee mug this morning and where your eyes are looking at.

On the other hand, the virtual and digital world will be experienced much more immersive. While users currently mainly interact with a two-dimensional interface that is mostly restricted to visual and auditive sense, the metaverse promises a three-dimensional interface and appealing to all senses.

This paper's view is: the metaverse is a spectrum of different digitally enhanced worlds converging physical and digital environments, not limited to virtual 3D environments

Combining those will enable a convergence of the internet and what is called Extended Reality (XR). It integrates digital and physical to various degrees and enables the blending of digital and physical existence via Augmented Reality (AR) and Virtual Reality (VR). Further information about XR can be found in the Technology Enablers section below.

For this to be viable at scale, the development of the metaverse must go through the following three sequential stages:

I) Digital Twins

This stage refers to large-scale and high-fidelity digital models and entities duplicated in virtual environments. The visual representation of those models is relevant, but just one of manifold factors. A digital twin requires all properties of their physical counterparts including objection motions, temperature and more. Their data will connect the two.

II) Digital Natives

Connected ecosystems, including culture, economy and regulations are at place to support the creation of objects within the digital world - some of which will be linked to physical counterparts.

III) Co-Existence of Physical-Virtual Reality

The name of this third and final stage speaks for itself. The virtual world will become self-sustaining and persistent. It co-exists and interoperates with the physical world. The 2030 vision outlined in the introduction of this paper describes an environment where this third step has been attained.

Undergoing the three stages of development and reaching a certain level of maturity will not happen overnight and is expected to take several years.

[4] Ning et al. (2021): A Survey on Metaverse: the State-of-the-art, Technologies, Applications, and Challenges
[5] Lee et al. (2021): All One Needs to Know about Metaverse: A Complete Survey on Technological Singularity, Virtual Ecosystem, and Research Agenda

This long time relates to both metaverse related soft- and hardware technology still being in development and personal and industrial adoption still being in infancy.

Technology Enablers

Actualizing the metaverse requires various key technologies that are all interrelated and depended on each other.[5]

With metaverse-related patents having doubled in the past five years, some of these technologies have already reached maturity allowing users to tap into what experiencing the metaverse might be like; however other technologies are still in development halting an outright roll out. [6]

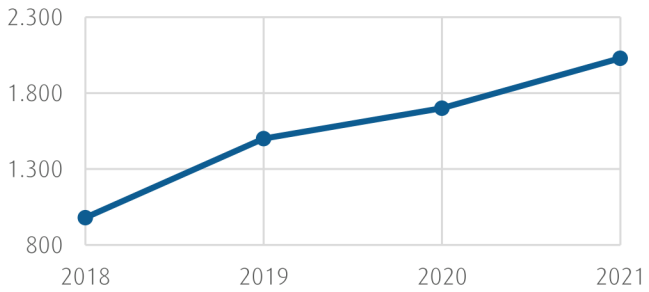


Figure 2: Evolution of metaverse-related patents (number of yearly patent submissions)

Extended Reality (XR)

Naturally, the central developing technology regards XR as it blends digital and physical existence and will allow users direct access to the metaverse.

The two main types of devices realizing such access are Virtual Reality (VR) and Augmented Reality (AR) devices. Both types speak to one's native human senses (including but not limited to the visual sense) with VR allowing users to obtain a realistic experience in a virtual world and AR focusing on the enhancement of the physical world and inter-operation with digital and real-life objects by superposing the virtual and physical environment.

Even though everyone trying out a VR headset will be surprised by its immersiveness, the long-term goal of achieving invisible computing interfaces conjointly speaking to all senses and enabling ubiquitous user interaction remains still far off in the future. Virtual reality sickness, which's symptoms include general discomfort, headache, stomach awareness, nausea, etc. appears when the array of users' senses do not harmonize/correlate. Minimizing those discrepancies is going to increase the level of immersion and will be achieved by overcoming the challenges of the technology challenges mentioned in this chapter.

Concrete obstacles for XR are fast and accurate human pose tracking, seamless and lightweight user interaction and improvements of focus blur.

Artificial Intelligence (AI)

First proposed in 1956, AI describes computer science applications whose goal is to exhibit intelligent behavior. In the metaverse context it serves two main functions:

First, it supports operators and designers by automizing laborious tasks and has through this proven to achieve higher performance than conventional approaches. In the Linde environment this converts e.g. to

[6] IFI Claims Patent Services (2022): Inventing the Metaverse. Technology companies are laying the patent groundwork for this whole new immersive 3D world.

the formation of high-fidelity digital twins, where - due to the plants' complexity and heterogenous data - deep learning is to be applied.

Second, it is utilized in computer vision and natural language processing, which allows computing devices to understand the information of the users' activities and their surroundings.

It is used for recognizing and understanding visual information of users' activities and their physical surroundings, such as real time hand position, gaze, emotion, speech, image, and text detection.

Nonetheless prevailing challenges include the interaction system needing to understand more complex environments, even more reliable and efficient body and pose tracking algorithms in addition to color correction, texture restoration and blur estimation. In parallel a challenge is development of more lightweight AI which requires less computing.

Internet of Things (IoT) and Robotics

Leveraging XR systems, IoT and robotics can invite human users to co-participate in data management and decision-making. Presenting abstract phenomena such as a data flow in easy-to-view manners will enable better human-in-the-loop decision making. In addition, robots navigating through an IoT-equipped industrial system can be utilized for remote repair work and the like.

Nonetheless, it is important to mention that metaverse-driven interaction with IoT and robots is still at a nascent stage and needs to be explored further to enable its full functionality in an industrial context.

IoT provides an opportunity to build powerful industrial systems by leveraging ubiquity of radio-frequency identification (RFID), and wireless, mobile, or sensor devices.

Network, Edge and Cloud

One of the most impactful factors on the immersive feeling is the delay between user input and its effect reflected on the display. This so-called latency needs to be below the human perceptible limit to allow seamless and direct interaction with the metaverse. Minimizing latency is critical for the metaverse; after all, a failure of which is more than just a small blemish: It results in immediate discomfort or even motion sickness and dizziness.

As of 2022, the primary latency bottleneck forms the last mile for wireless connection: Intensive computation required is too heavy for mobile devices. Offloading computation to the cloud is thus necessary to guarantee the timely processing and user experience in the long term. Current cloud distribution can deliver network latency of less than 100ms, but only a small minority of countries meet the threshold via wired networks and China is the single only country meet the threshold via wireless networks.

Consequently, until 5G and its successors are truly prevalent around the world, edge computing will be used as a complementary solution making it an essential pillar in the eyes of many industry insiders.

Key Players

The following corporations have been identified as the main players shaping metaverse-related developments at the time of writing this paper.

Meta Platforms

With Facebook's rebranding in late 2021, the company not only drew attention away from brand-related public criticism, but also drew massive public attention to the metaverse. This is discerned from Google's search trends presented in the graph below.

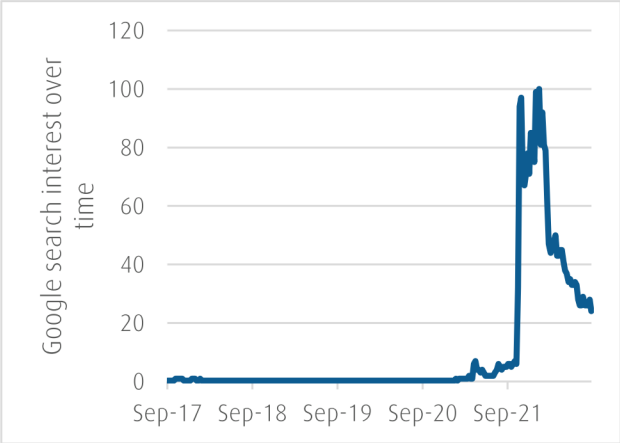


Figure 3: Worldwide Google Searches of the word Metaverse[7]

The rebrand coincided with a forward escape taking place in the company, since Facebook has seen a decrease of users for the first time in 2022.[8] The strategic target is shifting from a social media company to a platform company, providing the infrastructure for metaverse-related activities. Their model can be compared with that of Google (Play Store) or Apple (App Store) for mobile platforms. Meta announced to be expecting major returns from "in-app" commissions for transactions happening on their platform. At the time of writing this paper, Meta was heavily invested with one fifth of all Meta employees working on metaverse-related topics. Further, it is notable that Meta Platforms currently owns a 90% share of the VR devices market.
In terms of market focus, Meta addresses end users and B2C companies.

Microsoft

Microsoft's sprint into the metaverse is apparent on multiple divisions. Protecting their inventions in AI around human memory, virtual assistants in augmented reality and holographic augmentation, the company leads the metaverse-related patent space. They notably acquired the video game holding company Activision Blizzard for $70 billion, to utilize games such as Call of Duty as "building blocks for the metaverse".

One should not underestimate Microsoft's significance for professional organizations including Linde. Here the company offers two metaverse platforms: The "industrial metaverse" which is a platform for industry 4.0 and digital twin solutions. And the "Metaverse Workplace", which allows XR to be integrated directly into the existing Teams environment. Even at the time of writing this paper, users already can join colleagues' Teams meeting as a 3D avatar via a VR headset.[9] Microsoft further announced a collaboration with Meta Platforms, allowing Microsoft Office products to run on Meta VR devices and Meta's software platform.

[7] Google Trends (2022): Metaverse - Explore - Google Trends
[8] manager magazin (2022): Die hochriskante Meta-Wette des Mark Zuckerberg
[9] Microsoft (2022): Microsoft Mesh

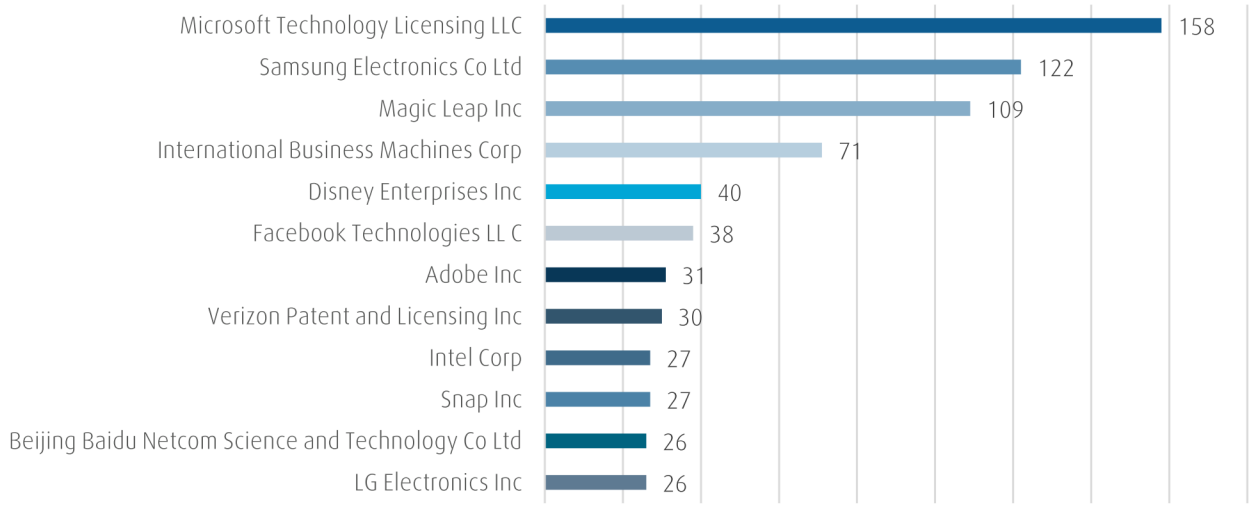


Figure 4: Top current patent applicants for metaverse technology[10]

Apple

Apple has not acknowledged publicly to any concrete plans regarding the metaverse. Nonetheless there is credible ground to expect them to play a major role in its establishment.

The company currently controls the largest AR platform encompassing soft- and hardware worldwide; the App Store with over 14,000 AR applications and millions of Apple's devices are AR-ready. The latter include LiDAR equipped iPhones and iPads and head position measuring AirPods which enable auditive AR. Factually without the burden of being the first to market, developments of both AR and VR wearables are taking place behind semi-closed doors. Analysts and Bloomberg have reported Apple plans to introduce an AR headset by this or next year with glasses to follow later.[11]

"Apple also did not invent the tablet. I tested my first tablet in 1992 when early pen-based tablets first became available. Eighteen years later, in 2010, Apple introduced the iPad, dramatically changing how we compute in mobile settings."

One can find a wide range of technologies that had - after existing for quite some time - been substantially improved by Apple and after integrating them into their ecosystem revolutionized entire markets. Examples of this include the mp3 player market and Apple's iPod, the tablet market and Apple's iPad, the smartphone market and Apple's iPhone and the smartwatch market and Apple's Apple Watch. And the same is expected to happen when Apple enters the AR & VR device market.[12]

[10] IFI Claims Patent Services (2022): Inventing the Metaverse. Technology companies are laying the patent groundwork for this whole new immersive 3D world.
[11] Reuters (2022): Apple teases metaverse AR plans, stock jumps
[12] Bajarin (2022): Could Apple Help Speed-Up The Adoption Of The Metaverse?

NVIDIA

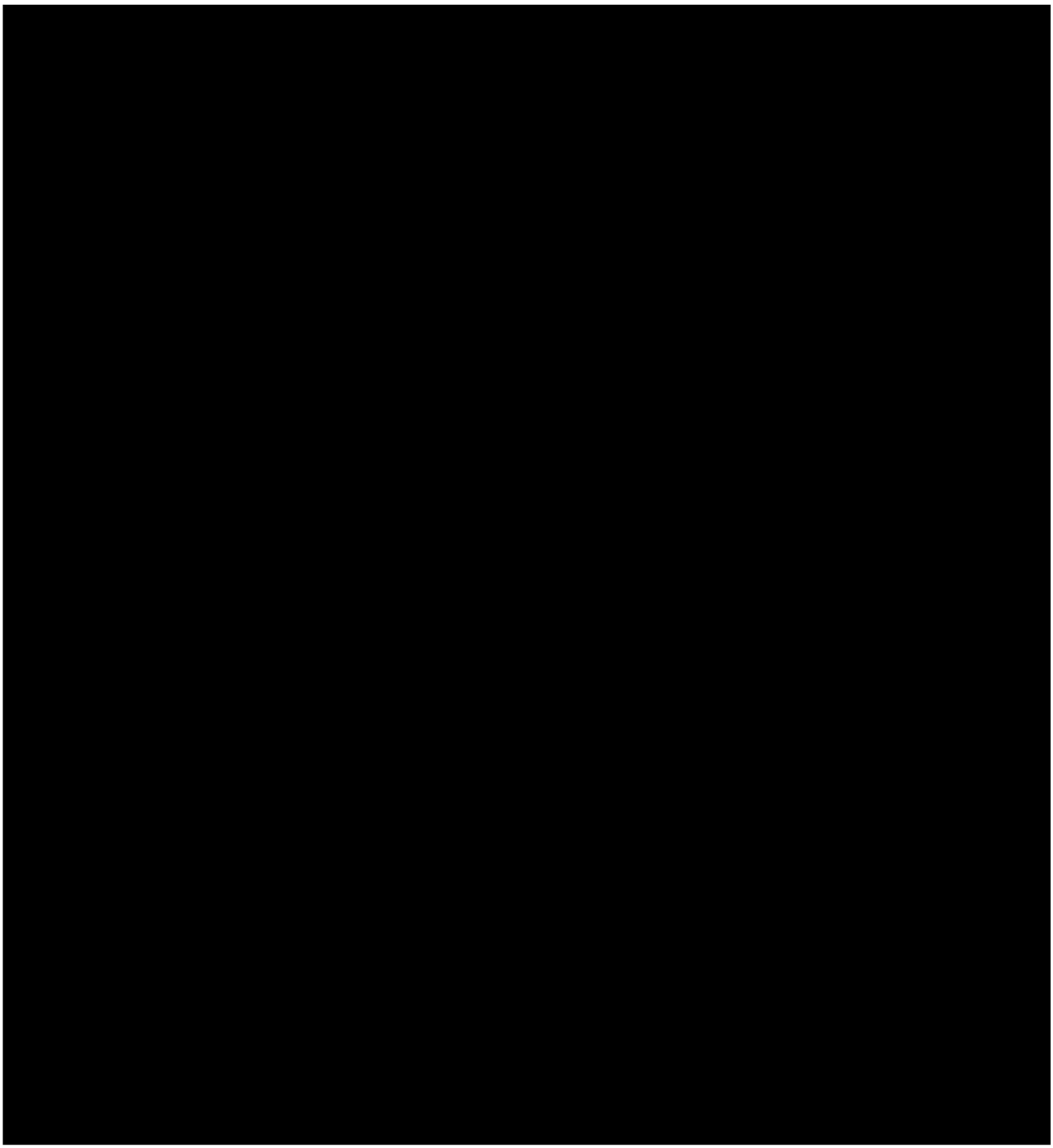
The California-based computer chip producer has launched a platform for building and operating XR applications: NVIDIA Omniverse Platforms. The end-to-end design, collaboration, and true-to-reality digital twin simulation platform enables individuals and corporations to connect various 3D worlds. It provides a toolset to create and link metaverses.[13]
With its approach of granting free licenses to artists and creatives, individual contributors are engaged to create 3D content which shall form the base for a scaling user ecosystem. On the B2B side, NVIDIA engaged in collaborations with numerous corporations supporting them in various digital twin initiatives. Omniverse users include Amazon, PepsiCo, BMW, and Siemens – all of which have announced to build their own metaverse content on the NVIDIA platform.

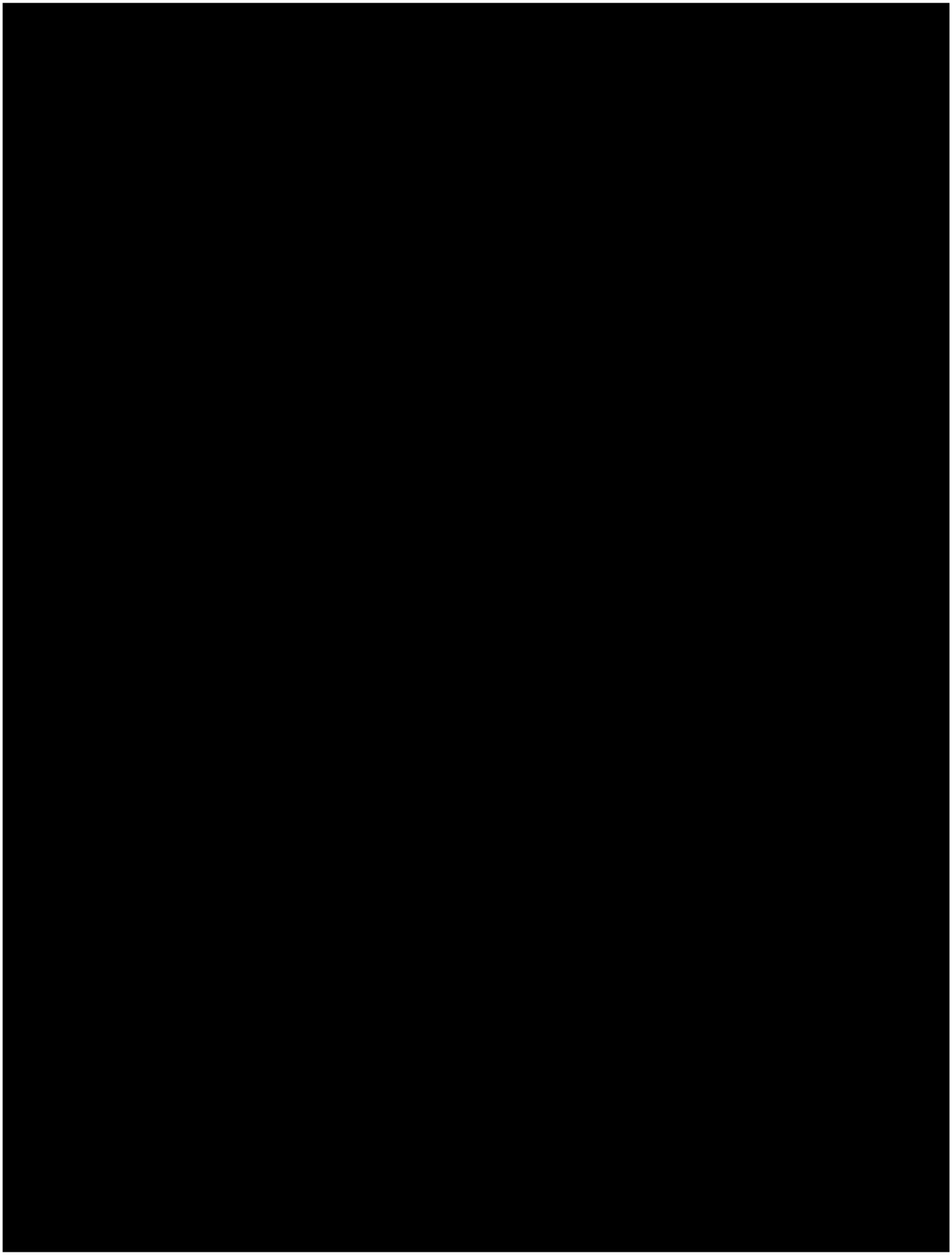
Others

As previously stated in this paper, there will not be one single metaverse, but there will be multiple ones. And so, besides the above-mentioned key players, already at the time of writing this paper, there are several metaverses (some even with millions of monthly active users), all with different user groups and use cases. As the development in the space is fast paced, also the key players shaping metaverse-related developments may change and respectively require continuous observation.

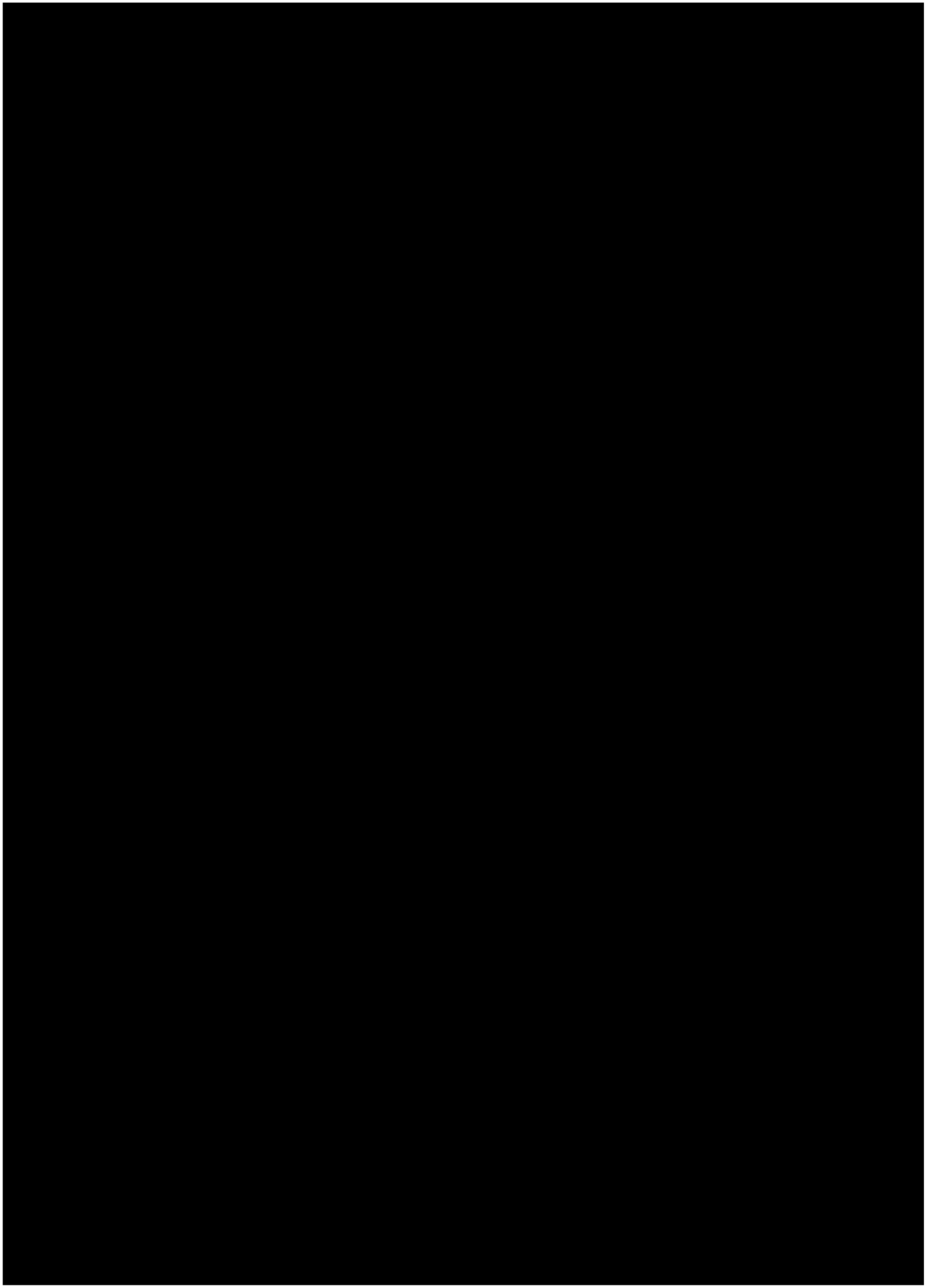
[13] NVIDIA (2021), The Metaverse Begins: NVIDIA Omniverse and a Future of Shared Worlds

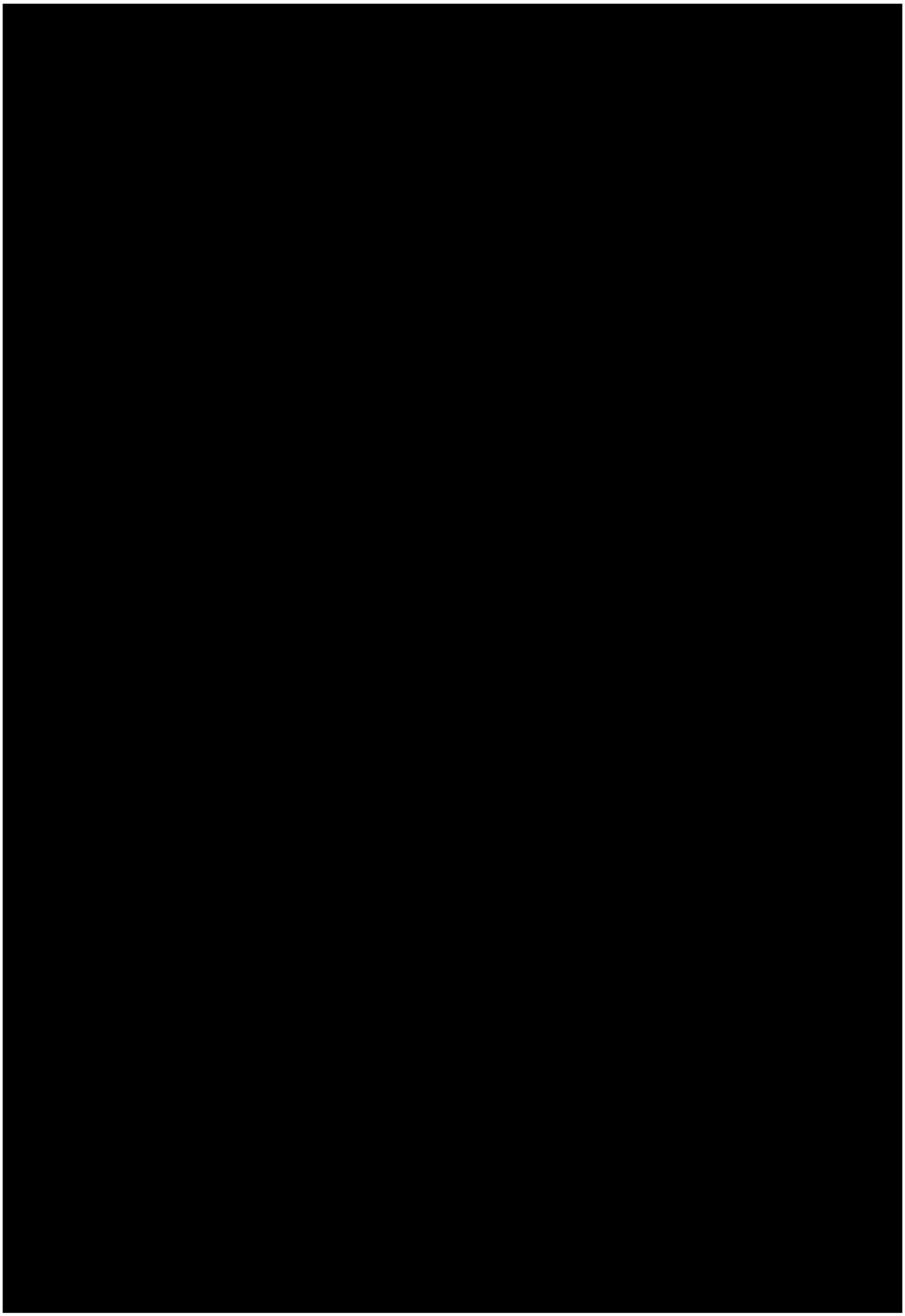
What does it mean for Linde? Use cases in perspective

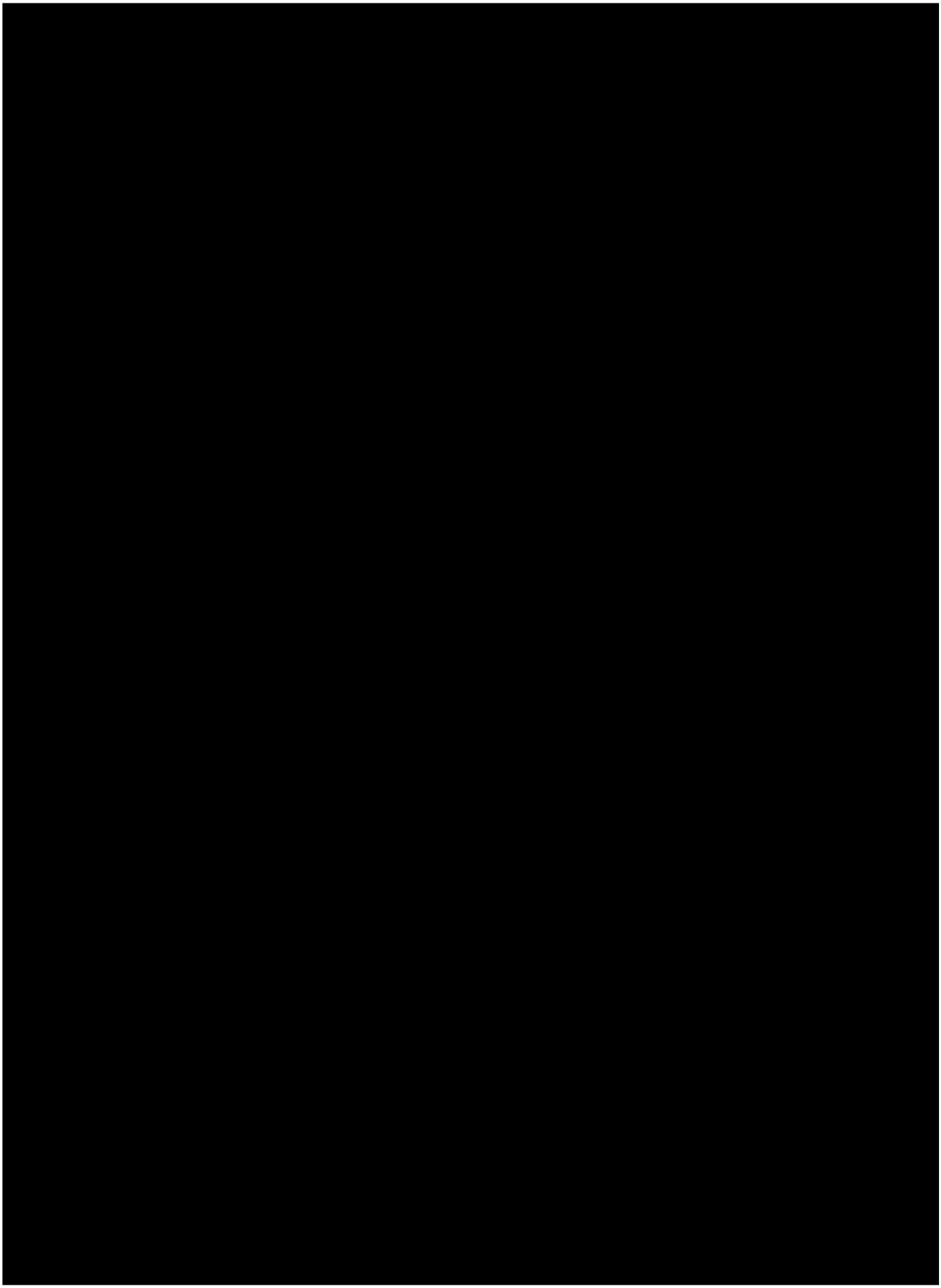




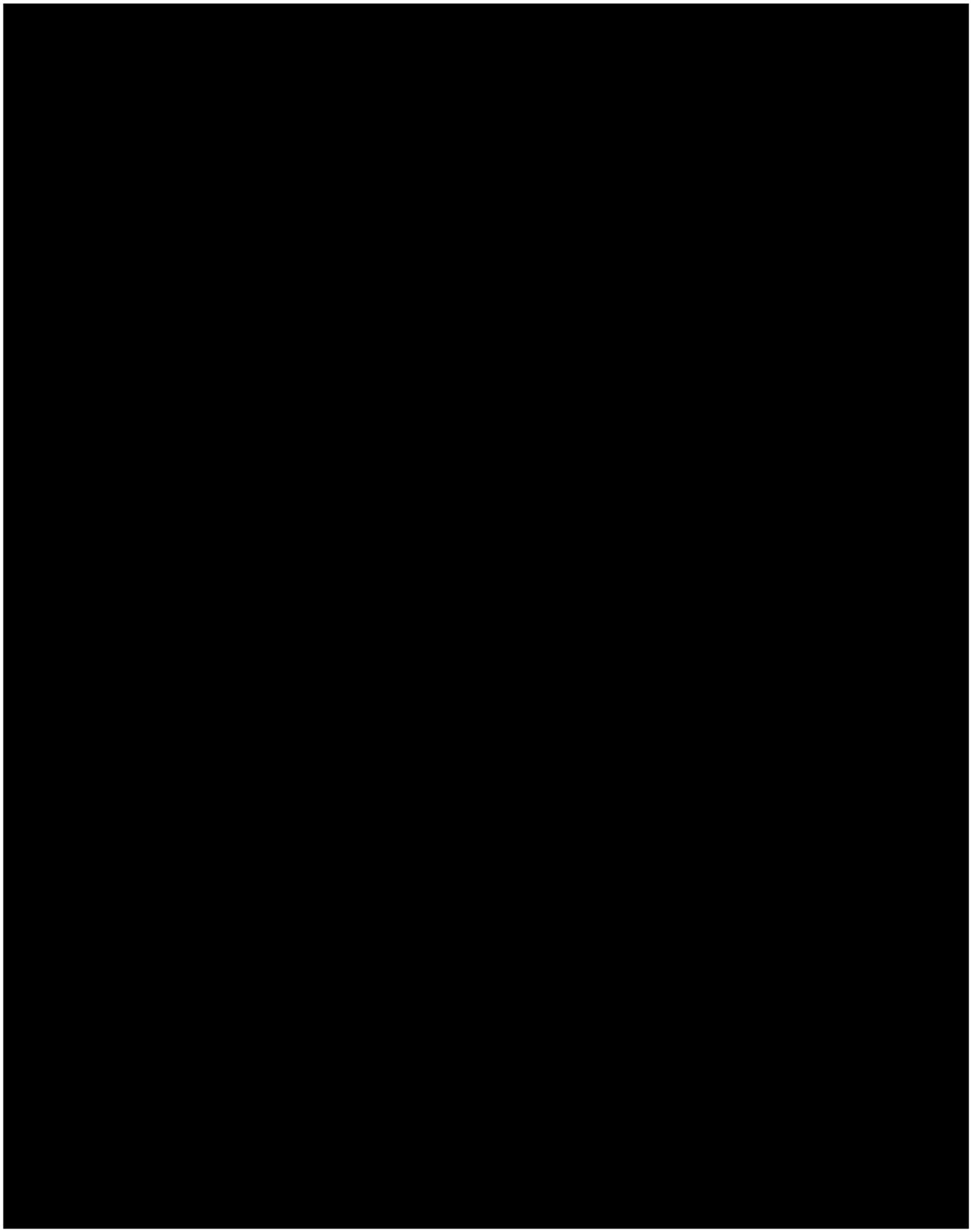
[14] Refer to https://vr.linde.com/



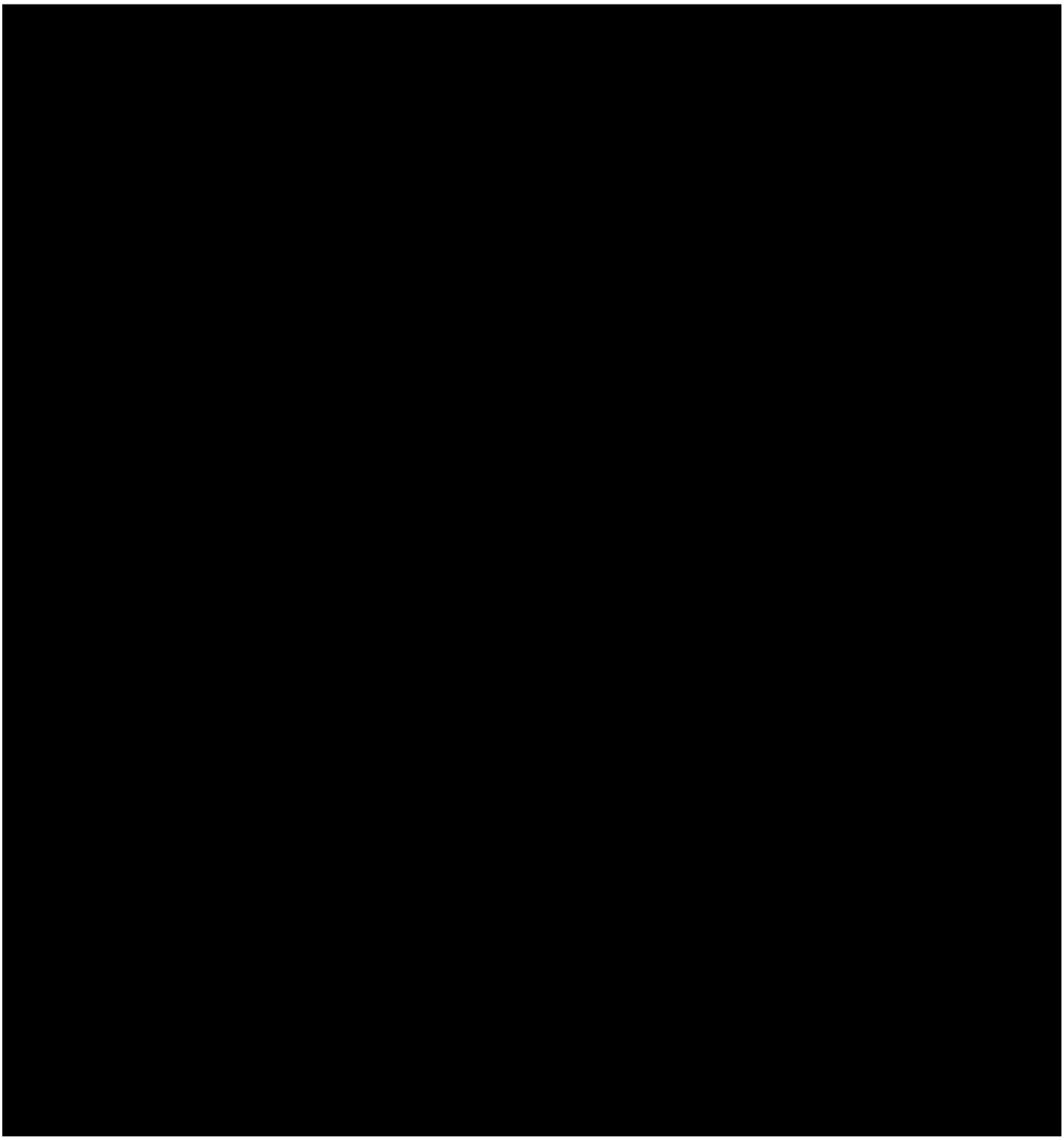




[15] Citi (2022): Metaverse and Money: Decrypting the Future, Citi Global Perspectives and Solutions



[16] Accenture (2022): Digital Health Technology Vision 2022, Meet Me in the Metaverse



[17] CNBC (2022), Chipmakers are set to be 'winners' as the metaverse takes off
[18] Microsoft (2022), Hydrogen fuel cells could provide emission free backup power at data centers

Conclusion

What will the metaverse be? And what will it do? Different players are already exploring ways it can deepen connectivity and complement everyday activities. And, within a decade, the metaverse has the potential to drive a very different world. While regarded as next evolution of the internet through integrating digital environments with the physical world, the metaverse will have significant impact on both, B2C applications, but also on B2B enterprise operations.

Even though most known use cases are directly consumer focusing (e. g. immersive 3D social networks) or supporting consumer focusing applications (e. g. consumer-app ecosystem infrastructures), various use cases already are – or will become relevant for industrial enterprises.

Even though the main use is consumer facing, it already becomes relevant for industrial enterprises.

